

NVH Source Locator —

NVH Source Locator —

NVH Source Locator                                     

- **Resolution:** The number of pixels that can be displayed horizontally and vertically. Resolution is measured in pixels (horizontal resolution and vertical resolution).

- **Refresh rate:** The number of times the screen is updated per second. Refresh rate is measured in Hertz (Hz) or frames per second (FPS).

Resolution and refresh rate are the two most important factors in determining the quality of a display.

Resolution and refresh rate are the two most important factors in determining the quality of a display. Resolution and refresh rate are the two most important factors in determining the quality of a display.

- 2 **Resolution** → Resolution is the number of pixels that can be displayed horizontally and vertically.

- 3 **Resolution** → Resolution is the number of pixels that can be displayed horizontally and vertically. 2D (X Y)

- 4 **Resolution** → Resolution is the number of pixels that can be displayed horizontally and vertically. 3D (X Y Z)

Resolution and Refresh Rate {#before-you-start}

Resolution and Refresh Rate:

- Resolution is the number of pixels that can be displayed horizontally and vertically. Resolution is measured in pixels (horizontal resolution and vertical resolution). Resolution is measured in pixels (horizontal resolution and vertical resolution).

- Resolution 2 Resolution is the number of pixels that can be displayed horizontally and vertically. Resolution is measured in pixels (horizontal resolution and vertical resolution). Resolution is measured in pixels (horizontal resolution and vertical resolution).

- Resolution Resolution is the number of pixels that can be displayed horizontally and vertically. Resolution is measured in pixels (horizontal resolution and vertical resolution).

- Resolution Resolution is the number of pixels that can be displayed horizontally and vertically. Resolution is measured in pixels (horizontal resolution and vertical resolution). Resolution is measured in pixels (horizontal resolution and vertical resolution).

**NVH Source Locator**

TDOA Calculator

PRO



2-Sensor

3-Sensor

3-Sen+

4-Sensor

4-Sen+

3D

3D+

Materials

Pair A-B

Pair A-C

Step 1

Calibration — Speed of Sound

Sensor spacing A-B

-

118

+

cm

External knock time delay

-

471

+

μs

Calculated speed of sound

2505 m/s

Closest: PEEK (2500 m/s)

+ Save as custom material

Step 2

Event Measurement

Event time delay

⊕ trials

-

227

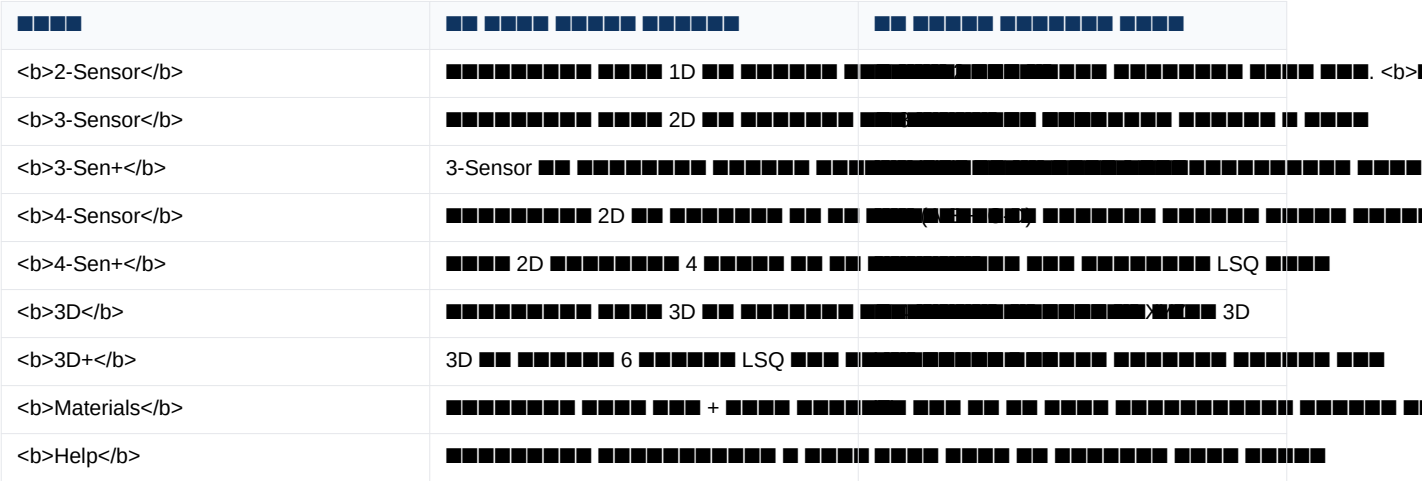
+

μs

Which sensor heard it first?

Sensor A

Calculate Source Location

[illegible]

XXXXXXXXXX XXXXXXXXXXXXXXXX: XXXXXXXXXXXXXX XXXX XX XXXXXXXX XXX XXX XX XXXXXXXXXXX.



NVH Source Locator
 TDOA Calculator PRO




2-Sensor 3-Sensor 3-Sen+ 4-Sensor 4-Sen+ 3D 3D+ Materials

Pair A-B

Pair A-C

Step 1

Calibration – Speed of Sound

Sensor spacing A-B

–

118

+

cm

External knock time delay

–

471

+

μs

Calculated speed of sound

2505 m/s

Closest: PEEK (2500 m/s)

+ Save as custom material

Step 2

Event Measurement

Event time delay

⊕ trials

–

227

+

μs

Which sensor heard it first?

Sensor A

Calculate Source Location

2-Sensor

1:

Materials . ("Mild (1020)". .

" " Mild (1020)". .

2

NVH

Source Locator

TDOA Calculator

PRO

2-Sensor

3-Sensor

3-Sen+

4-Sensor

4-Sen+

3D

3D+

Materials

Same triangle as 3-Sensor, but calibrate & measure **all three pairs** (A-B, A-C, B-C). The solver uses all 3 TDOAs in a least-squares fit – more robust to measurement noise and anisotropic materials. The *RMS residual* tells you how consistent your measurements are.

[▲ Show placement guide](#)

Setup

Triangle side lengths

A-B side length

-

100

+

cm

A-C side length

-

90

+

cm

B-C side length

-

80

+

cm

Pair A-B

Calibration & measurement

Calibration time delay – knock outside A-B

-

400

+

μs

2,500 m/s – Closest: PEEK (2,500 m/s)

Event time delay

-

120

+

μs

[⊕ trials](#)

Which sensor heard it first?

Sensor A

3-Sensor

[illegible][illegible]

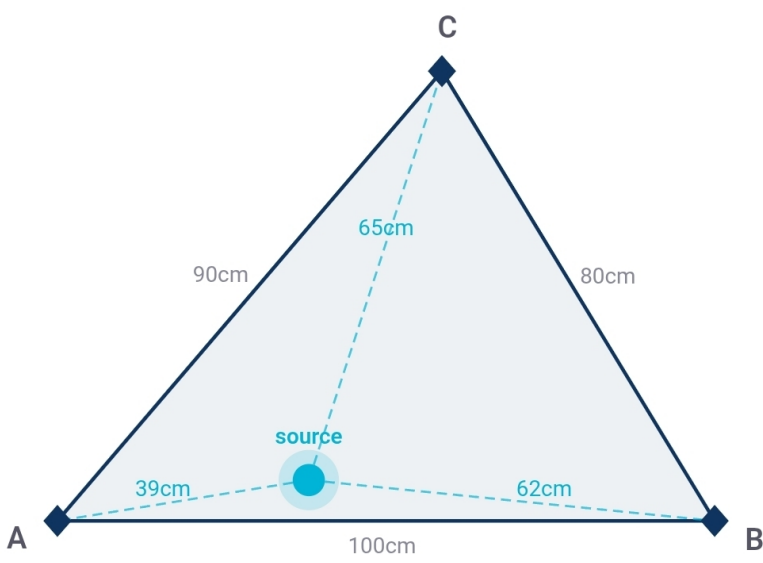
The following table shows the distances between the sensor corners (A-B, A-C, B-C) and the source.

The table also shows the estimated source location (X, Y) and the estimated source location (X, Y).

- tCal: The time taken for the signal to travel from the source to the sensor corners (X, Y)
- tEvent: The time taken for the signal to travel from the source to the sensor corners (X, Y)
- The estimated source location: (X, Y)

Sensor corners and source location

The diagram shows the sensor corners A, B, and C, and the source location X. The distances between the sensor corners and the source are given in the table below. The estimated source location (X, Y) is also shown.



A, B, C = sensor corners · dot = least-squares source estimate

Copy

Share

Print result

Annotate photo

The estimated source location is (X, Y).

1. 3-Sen+ (Pro)
 2. 4-Sensor
 3. 4-Sen+ (2D)
 4. 3D
 5. 3D+ (Pro)

3-Sen+ (Pro)

3-Sensor (A-B, A-C, B-C). 3 TDOA.

4-Sensor

- A-B = ()
- C-D = ()

A-B () C-D (). 2D.

4-Sen+ (2D)

(). A B C D.

3D

3D 4 (X Y Z). (A-B, A-C, A-D).

3D+ (Pro)

3D 6 (A F) LSQ. 3D.

Materials {#the-materials-tab}

20 °C.

■■■■ *Materials*

Page 10 of 10

[illegible][illegible]

14 **■■■■ ■■■■■■■■■■ ■■■■ ■■■■■■■■■■ ■■■■ ■■■■ ■■■■■■. ■■■■ ■■■■ ■■■■ ■■**
■■■■■■■■■ ■■ 20 °C ■■■■■■ ■■■■■■ ■■■■■■ ■■ ■■■■ ■■■■■■ ■■■■ ■■■■ ■■■■ ■■
■■■■■■■ ■■■■■■:

- ■■■■■■■■■■
- ■■■■■■■■ Mild (1020)
- ■■■■■■ ■■ ■■■■ (304)
- ■■■■ (■■■■■■■■■■)
- ■■■■
- ■■
- ■■■■
- ■■■■

- 
- 
- 
- 
- 
- 

1. 在 20 °C 下，将 1.0 g 的 1,4-二氯苯和 1.0 g 的 1,4-二氯苯加入到 10 mL 的 1,4-二氯苯中，搅拌均匀。

[illegible]

Figure 1. Schematic diagram of the in-situ monitoring system for the water level, water temperature, and water quality parameters. The system consists of a data logger, a water level sensor, a water temperature sensor, and a water quality sensor. The data logger is connected to the sensors and stores the data. The water level sensor is connected to the water level data logger. The water temperature sensor is connected to the water temperature data logger. The water quality sensor is connected to the water quality data logger.

5/5/2019

THE UNITED STATES OF AMERICA, by and through the undersigned, its duly authorized representative, hereby certifies that the foregoing is a true and correct copy of the original as the same appears in the records of the Department of State.

■■■■■ ■■■ {#temperature-compensation}

[illegible]

Page 10 of 10

■■■■■■■■ (■■■■■ ■) → ■■■■ ■■■■ ■■ ■■■ ■■■■. ■■■■ ■■■■ ■■■■■■ ■■■ ■■ ■■ °C
 ■■■■ ■■■■ (■■■■■■■ -40 ■■ +200).

- **Содержимое "Спецификации данных" не должно включать в себя никаких комментариев**
- **Спецификация данных должна быть написана на русском языке**
- **Спецификация данных должна быть написана в формате: "Температура: 60 °C"**

Спецификация данных

Спецификация данных должна быть написана на русском языке. Температура должна быть указана в °C. Спецификация данных должна быть написана в формате: "Температура: 60 °C".

Спецификация данных должна быть написана в формате: "Температура: 60 °C".

Спецификация данных

Спецификация данных должна быть написана на русском языке. Спецификация данных должна быть написана в формате: "ref only". Спецификация данных должна быть написана в формате: "ref only".

Спецификация данных {#photo-annotation}

Спецификация данных должна быть написана на русском языке. Спецификация данных должна быть написана в формате: "ref only".

 Save image



11 12 13 14 15 16 17 18 19

- Android:  PDF              
- iOS:                  

■■■■■■■■■■ ■■■■■■■■ **{#backup-and-restore}**

11 12 13 14 15 16 17 18 19 20 21

[illegible]

Page 10 of 10

■■■■■■■■ → ■■■■■■■■ → ■■■■ ■■■■■■■■ ■■ ■■ ■■■■■■■■ ■■■■ ■■■■■■■■ ■■■■. ■■■■■■■■
 ■■■■ ■■■■■■■■ ■■■■ ■■■■■■■■ ■■■■■■■■ ■ ■■■■■■■■ ■■ ■■■■ ■■■■■■■■.

THE INFORMATION CONTAINED HEREIN IS UNCLASSIFIED EXCEPT WHERE SHOWN OTHERWISE. THIS INFORMATION CONTAINS NEITHER RECOMMENDATIONS NOR CONCLUSIONS OF THE NATIONAL ARCHIVES. IT IS THE PROPERTY OF THE NATIONAL ARCHIVES AND IS LOANED TO YOUR AGENCY; IT AND ITS CONTENTS ARE NOT TO BE DISTRIBUTED OUTSIDE YOUR AGENCY.

■■■■■■■ {#settings}

THE UNITED STATES GOVERNMENT IS NOT RESPONSIBLE FOR THE CONTENTS OF THIS MESSAGE OR FOR ANY DAMAGE TO PROPERTY OR PERSONS THAT MAY BE CAUSED BY THE USE OF THE INFORMATION CONTAINED HEREIN.

Units & settings

×

LANGUAGE

English

DISTANCE

cm

inches

TIME

μs

ms

REFERENCE TEMPERATURE

Resets to 20 °C on next app launch

−

20

+

°C

Used for temperature-compensated material speeds. Always calibrate in-situ for best accuracy.

MEASUREMENT HISTORY

🕒 View recent calculations

⬇️ Backup

⬆️ Restore

REPORT

Report header text

e.g. EVDiag NVH Services · service@evdiag.net · +49-XXX-XXXXX

Appears at the top of every printed report.

0 / 500

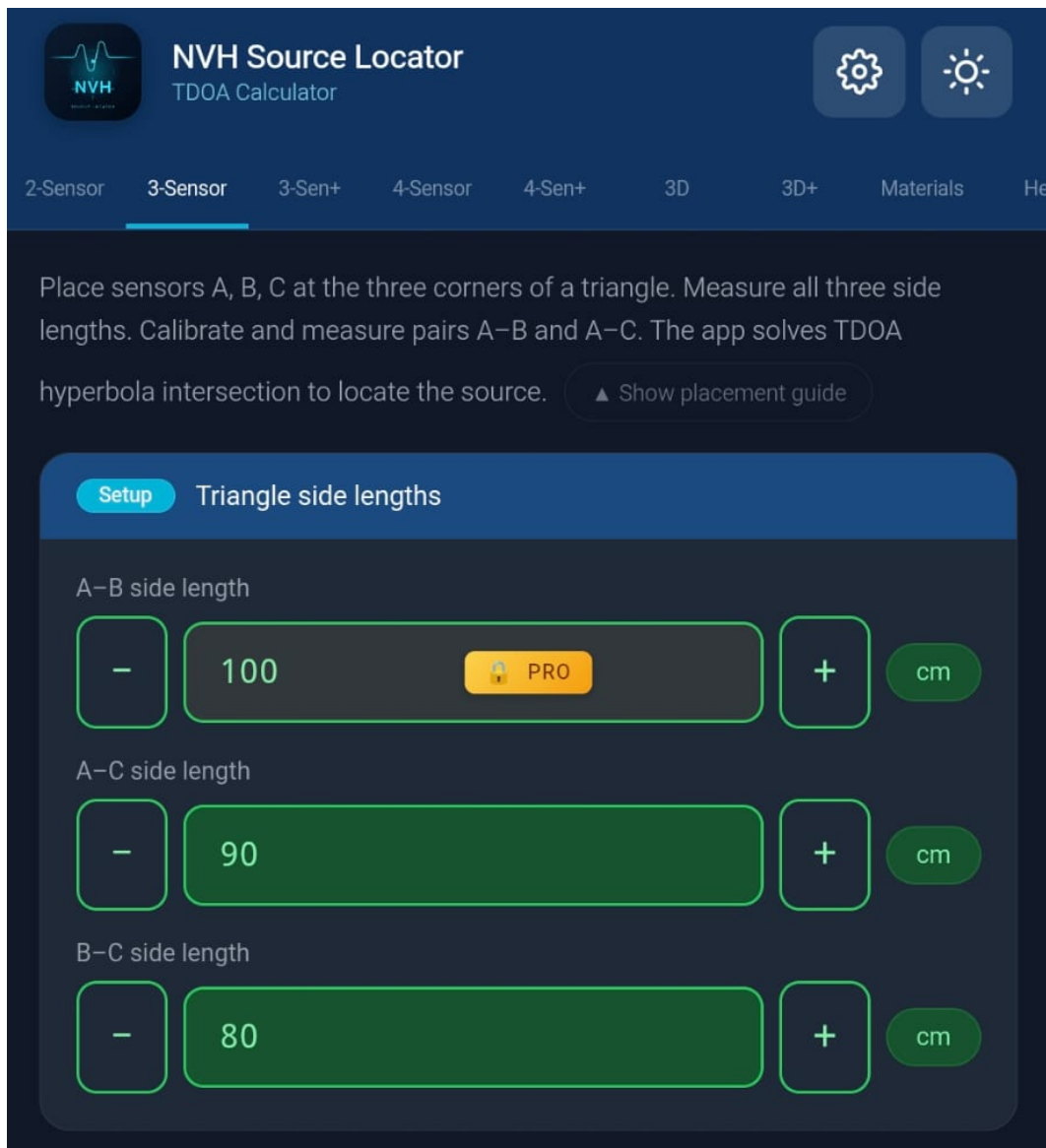
DATA

🕒 Show intro screen again

ABOUT

NVH Source Locator · TDOA Calculator for accelerometer-based source localization. Works offline. No data leaves your device.

[illegible]



paywall



Unlock NVH Pro

Full access to all features

- ✓ 3-Sensor & 4-Sensor triangulation
- ✓ 3D source location (X, Y, Z)
- ✓ Unlimited measurement history
- ✓ Save custom materials
- ✓ Backup and restore
- ✓ PDF reports with photo annotation
- ✓ Temperature compensation across all tabs

Unlock Pro — \$19.99

Have a Google Play promo code?

Looking for a discount code?
[Check the Picoscope Forum →](#)

Questions? support@evdiag.net

paywall

■■■■ ■■ ■■■■■ ■■■■■■■ ■■ ■■■■ ■■■ ■■■ ■■ ■■■ ■■■■■■■■ paywall ■■■■■■■ ■

■■■■ ■■■■■■:

- ■■■■ ■■■■■■■ ■■ ■■■■ PRO
- ■■■■ ■■■■■■■■
- ■■■■ ■■■ ■■■■■ ■■■ ■■ ■■■■ (\$19.99 ■■■■■■■■■ ■■■■■ ■■■ ■■■■ ■■■■■■■■■
■■■■■■■ ■■■■)

- **Download NVH Source Locator** (Android — iOS) **Apple Offer Code** **XXXXXXXXXXXX**)
- **Download NVH Source Locator** **Apple Offer Code** **XXXXXXXXXXXX**

Pro

Download NVH Source Locator **Pro** **Apple Offer Code** **XXXXXXXXXXXX**. **Download NVH Source Locator** **Pro** **Apple Offer Code** **XXXXXXXXXXXX** (Google Play **Android** **Apple App Store** **iOS**).

Pro

Download NVH Source Locator **Pro** **Apple Offer Code** **XXXXXXXXXXXX** (Google Play **Android** **Apple App Store** **iOS**):

- **Download NVH Source Locator** **Pro** **Apple Offer Code** **XXXXXXXXXXXX** (Google Play **Android** **Apple App Store** **iOS**)
- NVH Source Locator **Pro** **Apple Offer Code** **XXXXXXXXXXXX**
- **Download NVH Source Locator** **Pro** **Apple Offer Code** **XXXXXXXXXXXX** → **Download NVH Source Locator** **Pro** **Apple Offer Code** **XXXXXXXXXXXX**
- **Download NVH Source Locator** **Pro** **Apple Offer Code** **XXXXXXXXXXXX** **Pro** **Apple Offer Code** **XXXXXXXXXXXX**

Pro

Download NVH Source Locator **Pro** **Apple Offer Code** **XXXXXXXXXXXX** **Google Play Store** **App Store** **Apple Offer Code** **XXXXXXXXXXXX** **NVH Source Locator** **Pro** **Apple Offer Code** **XXXXXXXXXXXX** **Pro** **Apple Offer Code** **XXXXXXXXXXXX** — **Download NVH Source Locator** **Pro** **Apple Offer Code** **XXXXXXXXXXXX**.

Pro

Android: **Download NVH Source Locator** **Pro** **Apple Offer Code** **XXXXXXXXXXXX** **Google Play** **Apple Offer Code** **XXXXXXXXXXXX** **paywall** **Apple Offer Code** **XXXXXXXXXXXX** **Google Play** **Apple Offer Code** **XXXXXXXXXXXX**.

iOS: **Download NVH Source Locator** **Pro** **Apple Offer Code** **XXXXXXXXXXXX** **App Store** **Apple Offer Code** **XXXXXXXXXXXX** **Apple** **Apple Offer Code** **XXXXXXXXXXXX** **Google Play** **iOS** **Apple Offer Code** **XXXXXXXXXXXX** **Apple** **Apple Offer Code** **XXXXXXXXXXXX** **App Store** **Apple Offer Code** **XXXXXXXXXXXX**.

Help

Help **Apple Offer Code** **XXXXXXXXXXXX** **Apple Offer Code** **XXXXXXXXXXXX** **Apple Offer Code** **XXXXXXXXXXXX** **Apple Offer Code** **XXXXXXXXXXXX**.



Works offline

Installable on Android & iOS

No data sent anywhere



Single pair A-B or A-C. Linear result bar.



Two pairs, proportional
rectangle map.



Triangle mode, TDOA hyperbola solver.



49 materials. Tap to auto-fill calibration.



Mode explainers, placement guides, and a 10-item FAQ covering calibration, TDOA theory, and accuracy limits.

1

Strike the component outside the sensor pair. Enter the sensor spacing (cm) and the measured time delay (μs) – the app calculates the speed of sound in the material.

■■■■ Help

■■■■■■■■ ■■■■■ ■■■■■ ■■■■:

- [illegible]

- 在 20 °C 下，将 1.75 g 样品溶于 10 mL 水中。

故障排除 {#troubleshooting}

如果遇到问题，请按照以下步骤进行故障排除。

- 检查样品是否完全干燥。tCal 应在干燥条件下使用。如果样品潮湿，会导致结果不准确。 — 将样品在 20 °C 下干燥 24 小时。 — 在干燥过程中，应定期称重，直到重量恒定。
- 检查样品是否完全干燥。 — 将样品在 20 °C 下干燥 24 小时。 — 在干燥过程中，应定期称重，直到重量恒定。
- 检查样品是否完全干燥。 — 将样品在 20 °C 下干燥 24 小时。 — 在干燥过程中，应定期称重，直到重量恒定。

toast 将文件 "data.txt" 写入 "data.txt"

将文件 "data.txt" 写入 "data.txt"。如果文件不存在，将创建文件。

- 将文件 "data.txt" 写入 "data.txt"。
- 将文件 "data.txt" 写入 "data.txt"。
- 将文件 "data.txt" 写入 "data.txt"。

将文件 "data.txt" 写入 "data.txt"。

将文件 "data.txt" 写入 "data.txt"。如果文件不存在，将创建文件。 (速度 50 m/s 或 20,000 m/s)。 — 将文件 "data.txt" 写入 "data.txt"。

Materials 将文件 "data.txt" 写入 "data.txt"。

将文件 "data.txt" 写入 "data.txt"。如果文件不存在，将创建文件。 20 °C 将文件 "data.txt" 写入 "data.txt"。 — 将文件 "data.txt" 写入 "data.txt"。

将文件 "data.txt" 写入 "data.txt"。

将文件 "data.txt" 写入 "data.txt"。如果文件不存在，将创建文件。 1.75 g 将文件 "data.txt" 写入 "data.txt"。 — 将文件 "data.txt" 写入 "data.txt"。

将文件 "data.txt" 写入 "data.txt"。

将文件 "data.txt" 写入 "data.txt"。如果文件不存在，将创建文件。 20 °C 将文件 "data.txt" 写入 "data.txt"。 — 将文件 "data.txt" 写入 "data.txt"。

将文件 "data.txt" 写入 "data.txt"。

将文件 "data.txt" 写入 "data.txt"。如果文件不存在，将创建文件。 — 将文件 "data.txt" 写入 "data.txt"。

